Experiment 2: Design and Implementation of Sum of n Natural Numbers Algorithm using Verilog

Objective

The objectives of this experiment are:

- To design, implement, and verify a Verilog HDL module that computes the sum of the first n natural numbers, defined as:

$$S = \sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

- To understand the importance of datapath and control logic in digital system design.

Block Diagram

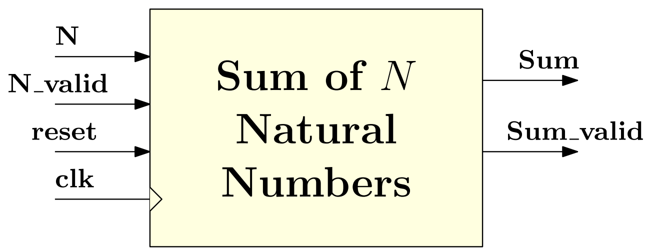

Figure 1: Block diagram of Sum of N Natural Numbers

Experiment Procedure

1. Develop the algorithm pseudocode and datapath/control architecture.
2. Implement and verify the given designs (Questions 1–6) using parameterized Verilog HDL code.
3. Comment on the area implications of the different design approaches.

Design Questions

1. Design a datapath-only Verilog module to compute the sum of the first N natural numbers using up-counting logic, where an index variable i increments from 1 to N and the sum is accumulated as $sum = sum + i$. Identify and explain the corner cases.

2. Enhance the datapath from Question 1 by incorporating a two-state finite state machine (e.g., `IDLE`, `BUSY`) to eliminate the identified corner cases and ensure controlled operation.

3. Further enhance the design by implementing a three-state FSM (e.g., `IDLE`, `BUSY`, `DONE`) to improve control flexibility and support system-level integration.

4. Modify the summation approach used in Question 1 by adopting a down-counting strategy, where the index variable i is initialized to N and decremented from $i = N$ to $i = 1$ with the three-state FSM implemented in Question 3.

5. Enhance the design from Question 4 by incorporating an acknowledgment (`ACK`) signal from a downstream module. The design shall assert completion in the `DONE` state and wait for `ACK` before returning to the `IDLE` state.

6. Restructure the implementation by separating the datapath and control logic into independent modules and instantiating them in a top-level module.